



Pranav Khare &lt;khare.pranav@gmail.com&gt;

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**Invitation to Review for IEEE Access Access-2024-13258**

1 message

**IEEE Access** <onbehalf@manuscriptcentral.com>

Thu, May 16, 2024 at 2:36 AM

Reply-To: ieeeaccess2@ieee.org

To: khare.pranav@gmail.com

16-May-2024

Dear Mr. Khare:

The manuscript entitled "Reinforcement Learning Assisted Adaptive Client Parameter Updates for Federated Learning" has been submitted to IEEE Access. Since your expertise is relevant to this article, we would be grateful if you would consider reviewing it.

The abstract appears at the end of this letter, along with the names of the authors. Please let me know within 2 days if you will be able to accept my invitation to review. If you agree to review, we ask that you provide your review comments within **7 days** to assist IEEE Access in maintaining its expedited peer review process of 4 to 6 weeks. If you are interested in reviewing but need additional time to complete the review, we can grant a reasonable deadline extension upon request. For additional information on IEEE Access and tips on completing a quality review, please visit our [Reviewer Best Practices guide](#).

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Thank you for your assistance with the peer review of this IEEE Access article.

Sincerely,

Prof. Derek Abbott  
Editor-in-Chief, IEEE Access

[ieeeaccess2@ieee.org](mailto:ieeeaccess2@ieee.org)

## MANUSCRIPT DETAILS

TYPE: Research Article

TITLE: Reinforcement Learning Assisted Adaptive Client Parameter Updates for Federated Learning

AUTHORS: PAN, ZIROU; WEI, LINNA; Zhao, Wei

**ABSTRACT:** Federated learning is widely applied in wireless communication, mobile technologies and daily life. It allows multiple parties to jointly train a model on their combined data without revealing any of their local data to a centralized server. However, in practical applications, federated learning requires frequent communication between clients and servers, which brings a considerable burden. In this work, we propose a method federated learning deep Q-network (FLDQN) to reduce the communication frequency between clients and servers in federated learning. FLDQN selects the parameter-update-frequency of a client adaptively and finds the best trade-off between local update and global parameter aggregation. Adaptive client's parameter-update-frequency selection mechanism in federated learning can be modeled as a Markov Decision Process in which we do not need any prior knowledge of the environment. We then propose a client's parameter-update-frequency selection scheme based on reinforcement learning that learns to effectively used the heterogeneous client resources in each communication round. We use Fashion-MNIST and CIFAR-10 to test FLDQN. The experimentation results show that FLDQN is superior to the benchmark schemes in terms of learning accuracy, convergence rate and the frequency of global aggregation. FLDQN also effectively reduces the communication overhead among the nodes in our experiments.

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